Large-Scale Multiple Inequality Testing without Data Snooping Bias

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Lecture Outline

1. Introduction
2. Tests that Control FWER
   - One-Step Tests: RC and SPA
   - Stepwise Tests: Step-RC and Step-SPA
   - Asymptotic Properties
   - Generalized Stepwise Tests: Step-SPA(k)
   - Empirical Applications
3. Tests that Control FDR
   - The Benjamini-Hochberg Procedure
   - The Storey Procedure
4. Concluding Remarks

A Multiple Testing Problem

In practice, it is common to find a superior model (trading rule, mutual fund) among a collection of models. This is a multiple testing problem because it requires testing many hypotheses (models).

- Testing model performance based on many individual tests may be misleading. When m statistics (each at 5% level) are independent, the probability of at least one false rejection is $1 - 0.95^m$:

m	1	2	5	10	50	100
Prob	0.050	0.098	0.226	0.401	0.923	0.994

- "given enough computer time, we are sure that we can find a mechanical trading rule which 'works' on a table of random numbers" (Jensen and Benington, 1970, *J. of Finance*, **25**, p. 470).

Familywise Error Rate

Let $d_{k,t}$, $k = 1, \ldots, m$ and $t = 1, \ldots, n$, be performance measures of the k-th model (relative to a benchmark) at time t and $\mathbb{E}(d_{k,t}) = \mu_k$ for all t.

Multiple Inequality Testing

Multiple null hypotheses are:

$$H_0^k : \mu_k \leq 0, \quad k = 1, ..., m.$$

For this testing problem, one must determine an overall type-I error probability for this family of hypotheses. Letting $\mathcal{I}$ denote the set of indices of true null hypotheses, one error probability is:

$$\text{Familywise error rate (FWER)} = \mathbb{P}_{H_0}(\text{Reject at least one } H_0^k,\ k \in \mathcal{I}).$$

Some Challenges

To control FWER in this multiple testing problem, there are some practical challenges:

- $d_{k,t}$ depend on the same data information and hence are highly correlated, e.g., the returns of different mutual funds all depend on the same market index; this causes the so-called data snooping bias (Lo & MacKinglay, 1990).
- The number of hypotheses, m, is large.
- H_0^k are inequalities: How should the null distribution be determined?

Individual Tests

To control FWER at α, one may set each individual test at the level of α/m. By the Bonferroni inequality,

$$\text{FWER} \leq \sum_{k \in \mathcal{I}} \mathbb{P}(\text{Reject } H_0^k) \leq \sum_{k \in \mathcal{I}} \frac{\alpha}{m} < \frac{m_o}{m}\alpha < \alpha,$$

where m_o is the number of true null hypotheses.

Holm (1979) proposes a stepwise procedure that checks the ordered p-values: $p_{(1)} < p_{(2)} < \cdots < p_{(m)}$ and rejects the hypotheses with

$$p_{(i)} < \frac{\alpha}{m - i + 1}.$$

As the threshold $\alpha/(m - i + 1)$ increases with i, Holm's procedure in effect permits more rejections. Yet, both tests would be too conservative to reject the null when m is large.

Joint Test

Let $\boldsymbol{d}_t = (d_{1,t}, \ldots, d_{m,t})^\top$ with $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{d}_t)$ and $\bar{\boldsymbol{d}}_n = n^{-1}\sum_{t=1}^n \boldsymbol{d}_t$. One may consider a joint test of $\boldsymbol{\mu}$ based on asymptotic normality:

$$\sqrt{n}(\bar{\boldsymbol{d}}_n - \boldsymbol{\mu}) \xrightarrow{D} \mathcal{N}(\mathbf{0},\ \boldsymbol{\Omega}),$$

where $\boldsymbol{\Omega}$ is the asymptotic covariance matrix.

- It is difficult to determine the null distribution when H_0^k are inequalities.
- Even when the null distribution is determined, implementing a joint test is not easy, especially when m is large and $d_{k,t}$ are closely related; for example, consistent estimation of $\boldsymbol{\Omega}$ is practically cumbersome.

White's Reality Check (RC)

- RC of White (2000) is a maximal statistic with the null distribution determined by the least favorable configuration (LFC), i.e., $\boldsymbol{\mu} = \mathbf{0}$,

$$\mathrm{RC}_n = \max_{k=1,\ldots,m} \sqrt{n}\bar{d}_k \xrightarrow{D} \max_{i=1,\ldots,m} Z_i,$$

where $(Z_1, \ldots, Z_m) \sim \mathcal{N}(\mathbf{0}, \ \boldsymbol{\Omega})$. Due to the LFC, White's RC is, by construction, a conservative test.

- The inference is based on the bootstrapped null distribution:

$$\max_{k=1,\ldots,m} \sqrt{n}\big(\bar{d}_k^*(b) - \bar{d}_k\big), \quad b = 1, \ldots, B,$$

where $\bar{d}_k^*(b)$ is the sample average of the b-th bootstrapped sample.
Note: Bootstrap the entire vector $\boldsymbol{d}_t = (d_{1,t}, d_{2,t}, \ldots, d_{m,t})^\top$, $t = 1, \ldots, n$, to preserve dependence across $d_{k,t}$.

Least Favorable Configuration

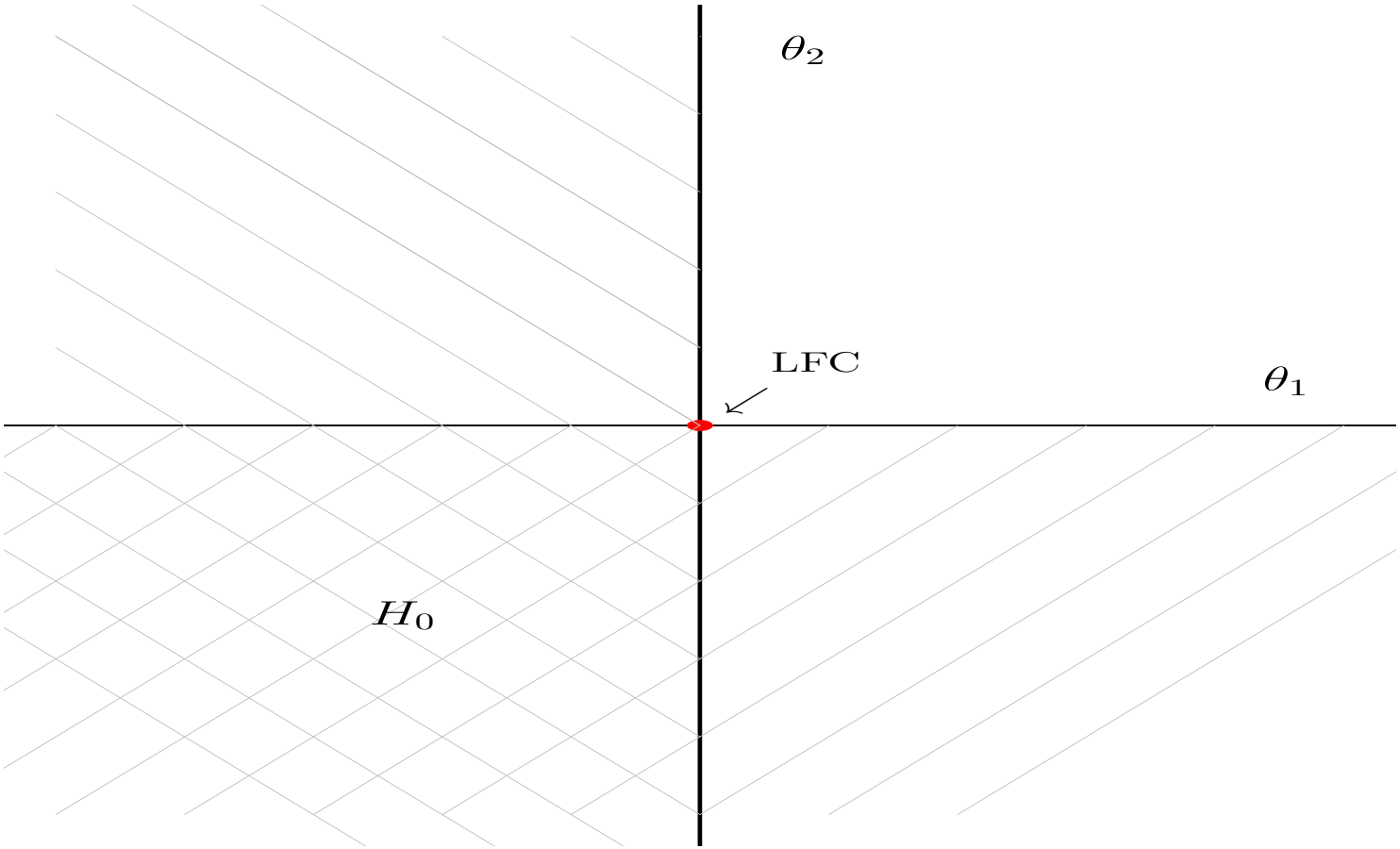


Figure: The configuration under the null ($\theta_1 \leq 0$ nad $\theta_2 \leq 0$) that is least favorable to the alternative.

Drawbacks of White's RC

When $\mu_1 = \ldots = \mu_\ell = 0$ and some $\mu_j < 0$ $(j = \ell + 1, \ldots, m)$,

$$\sqrt{n}(\bar{d}_1, \ldots, \bar{d}_\ell)^\top \xrightarrow{D} \mathcal{N}(\mathbf{0},\ \mathbf{\Omega}_o),$$

where $\mathbf{\Omega}_o$ is a sub-matrix of $\mathbf{\Omega}$, and

$$\sqrt{n}\bar{d}_j = \sqrt{n}(\bar{d}_j - \mu_j) + \sqrt{n}\mu_j \xrightarrow{\mathbb{P}} -\infty, \quad j = \ell + 1, \ldots, m.$$

This shows that, when LFC fails,

$$\mathrm{RC}_n \xrightarrow{D} \max_{i=1,\ldots,\ell} Z_i,$$

so that the limiting distribution does not depend on the "poor" models with negative means.

The limiting distribution of $\max_{i=1,\ldots,\ell} Z_i$ is not readily obtained because we do not know ℓ, the number of $\mu_j = 0$. White's RC bootstraps the distribution of $\max_{i=1,\ldots,m} Z_i$, and

$$\max_{i=1,\ldots,\ell} Z_i \leq \max_{i=1,\ldots,m} Z_i.$$

As such, the true distribution (LHS) is stochastically dominated by that of the RHS, and the bootstrapped critical values are larger than they should be. In other words, when LFC is not satisfied, the power of RC would deteriorate when more "poor" models with $\mu_j < 0$ are included in the test. Indeed, the power of RC could be driven to zero by including many poor models (Hansen, 2005).

Hansen's Superior Predictive Ability (SPA) Test

The SPA statistic of Hansen (2005), $\text{SPA}_n = \max\big(\max_{k=1,\ldots,m} \sqrt{n}\bar{d}_k,\ 0\big)$, relies on re-centered, bootstrapped distribution:

$$\max_{k=1,\ldots,m} \sqrt{n}\big(\bar{d}_k^*(b) - \bar{d}_k + \hat{\mu}_k\big), \quad b = 1, \ldots, B,$$

where $\hat{\mu}_k = \bar{d}_k \mathbf{1}\big\{\sqrt{n}\bar{d}_k \le A_{n,k}\big\}$ and $A_{n,k} = -\hat{\sigma}_k\sqrt{2\log(\log n)}$, with $\hat{\sigma}_k^2$ the estimate of the k-th diagonal element of $\mathbf{\Omega}$.

- When $\mu_k = 0$, $\hat{\mu}_k = 0$ almost surely.
- When $\mu_k < 0$, $n^{1/2}\bar{d}_k \le A_{n,k}$ with probability approaching one, so that $\hat{\mu}_k \xrightarrow{\mathbb{P}} \mu_k$.
- Re-centering leads to a better approximation to the distribution of $\max_{i=1,\ldots,\ell} Z_i$; the SPA test is thus more powerful than RC.

Studentization

One may employ a studentized statistic: $\max\left(\max_{k=1,\ldots,m} \sqrt{n}\bar{d}_k/\hat{\sigma}_k,\ 0\right)$,
The re-centered, bootstrapped distribution is then computed as:

$$\max_{k=1,\ldots,m} \sqrt{n}\left(\frac{\bar{d}_k^*(b)}{\hat{\sigma}_k^*(b)} - \frac{\bar{d}_k}{\hat{\sigma}_k} + \hat{\mu}_k\right), \quad b = 1, \ldots, B,$$

where $\hat{\sigma}_k^*(b)$ is the b-th boostrapped standard error,

$$\hat{\mu}_k = \frac{\bar{d}_k}{\hat{\sigma}_k}\mathbf{1}\left\{\sqrt{n}\bar{d}_k \leq -\hat{\sigma}_k\sqrt{2\log(\log n)}\right\}.$$

The studentized statistic usually results in better power performance than does the non-studentized version.

Note: We may replace $\log(\log n)$ in $A_{n,k}$ by some $a_n \to \infty$ and $a_n/n \to 0$. For example, one may use $\log n$ instead.

Stepwise RC and SPA Tests

When RC/SPA rejects, we know there is at least one outperforming model but do not know what they are. Step-RC of Romano and Wolf (2005) and Step-SPA of Hsu, Hsu, and Kuan (2010) are stepwise procedures that identify as many outperforming models as possible, while controlling

$$\text{FWER} = \mathbb{P}_{H_0}(\text{Reject at least one } H_0^i,\ i \in \mathcal{I}),$$

where $\mathcal{I}$ is the set of indices of true null hypotheses.

1. Reject model k when $n^{1/2}\bar{d}_k$ is greater than the bootstrapped RC/SPA critical value.
2. Remove $\bar{d}_k$ of the rejected models from the data and re-bootstrap the critical value using the remaining $\bar{d}_j$.
3. Repeating (1) and (2) based on the newly bootstrapped critical value.

The procedure stops when no model can be rejected.

Asymptotic Properties

Step-SPA Test (Hsu, Hsu, and Kuan, 2010)

1. (Exact FWER) Given level α_0, the Step-SPA test has FWER $= \alpha_0$ when n tends to infinity if and only if there is at least one $\mu_k = 0$.
2. (Consistency) H_0^k with $\mu_k > 0$ will be rejected by the Step-SPA test with probability approaching 1 when n tends to infinity.
3. (Power) The Step-SPA test is more powerful than the Step-RC test under the notions of power defined in Romano and Wolf (2005).

- The FWER of the Step-RC test $\leq \alpha_0$.
- If there is no $\mu_k = 0$, the FWER would be zero asymptotically, so that no null hypothesis will be incorrectly rejected.

The power notions in Romano and Wolf (2005):

- Minimal power: Probability of rejecting at least one false null hypothesis.
- Global power: Probability of rejecting all false null hypotheses.
- Average power: The average of the individual probabilities of rejecting each false null hypothesis. This is equivalent to the expected number of false hypotheses that will be rejected.
- The expected proportion of false hypotheses that will be rejected.
- The probability of rejecting at least $\gamma \times 100\%$ of the false null hypotheses, where γ is a user-chosen parameter.

Generalized Familywise Error Rate

FWER is a rather stringent criterion:

$$\text{FWER} = \mathbb{P}_{H_0}(\text{Reject at least one } H_0^i,\ i \in \mathcal{I}).$$

When m is large, only one false rejection is not a very serious error. This suggests that one may tolerate several incorrect rejections and relax the FWER control accordingly. The resulting test is capable of identifying more superior models. The generalized FWER of Lehmann and Romano (2005) is, for a pre-specified k,

$$\text{FWER}(k) = \mathbb{P}_{H_0}(\text{Reject at least } k \text{ of } H_0^i,\ i \in \mathcal{I}).$$

That is, false rejections would not be considered as an error unless there are k or more false rejections.

Step-SPA(k) Test

Step-RC(k) of Romano and Wolf (2007) and Step-SPA(k) of Hsu, Kuan, and Yen (2014) both achieve the asymptotic control of FWER(k).

- Let k-max$\{\mathcal{Y}\}$ denote the k-th largest value of the numbers in $\mathcal{Y}$.
- We bootstrap $\widehat{T}_j = \bar{d}_j/\hat{\sigma}_j$ and obtain $\psi_j(b) = \widehat{T}_i^*(b) - \widehat{T}_j$. For any subset $K \subseteq \{1, \ldots, m\}$, the critical values $q_K(\alpha, k)$ are determined as the α-th quantile of $k\text{-max}\left\{\psi_j(b) + \sqrt{n}\hat{\mu}_j : j \in K\right\}$ and $\hat{q}_K(\alpha, k) = \max\{q_K(\alpha, k),\ 0\}$.
- At each step, the procedure stops when $s < k$ models are rejected. If $s \geq k$ models are rejected, we do not simply bootstrap the remaining statistics because some (at most $k-1$) of these s models might have been incorrectly rejected. This possibility must be taken into account in bootstrapping the critical values in the next step.

Step-SPA(k) Algorithm

1. Reject H_0^ℓ if $\sqrt{n}\widehat{T}_\ell > \hat{w}_{A_1}(1-\alpha_0, k) = \hat{q}_{A_1}(1-\alpha_0, k)$, where $A_1 = \{1, \ldots, m\}$; otherwise, the procedure stops.
2. Let R_2 be the collection of the indices ℓ of the rejected hypotheses H_0^ℓ in the previous step and A_2 the collection of the indices of the remaining hypotheses. If $|R_2| < k$, then stop; otherwise, let

$$\hat{w}_{A_2}(1-\alpha_0, k) = \max_{R_2^\dagger \subset R_2,\ |R_2^\dagger| = k-1} \left\{ \hat{q}_K(1-\alpha_0, k) : K = A_2 \cup R_2^\dagger \right\}.$$

   That is, we bootstrap the critical values based on $k-1$ rejected $\widehat{T}_i$ for $i \in R_2^\dagger$ and $\widehat{T}_j$ for $j \in A_2$. We then reject H_0^ℓ with $\ell \in A_2$ if $\sqrt{n}\widehat{T}_\ell > \hat{w}_{A_2}(1-\alpha_0, k)$.
3. Repeat the previous step (with R_2 and A_2 replaced by R_3 and A_3, R_4 and A_4 and so on) till there is no more than k rejections.

Asymptotic Properties

Step-SPA(k) Test (Hsu, Kuan, and Yen, 2014)

1. (FWER(k)) Given the pre-specified level α_0, the asymptotic FWER(k) of Step-SPA(k) will be less than or equal to α_0.
2. (Consistency) The hypothesis H_0^ℓ with $\mu_\ell > 0$ will be rejected by Step-SPA(k) with probability approaching one.
3. (Power) Step-SPA(k) is more powerful than Step-RC(k) under the notions of power defined in Romano and Wolf (2005).

- Step-SPA(k) rejects whenever Step-RC(k) rejects but not conversely.
- The critical value in the last step of Step-SPA(k) is no greater than that of Step-SPA. Thus, all models rejected by Step-SPA must be rejected by Step-SPA(k) but not conversely.

Empirical Applications

1. Technical trading rule: Sullivan, Timmermann, and White (1999), White (2000), Hsu and Kuan (2005), Qi and Wu (2006), Hsu, Hsu, and Kuan (2010), Shynkevich (2012), Kuang, Schröder and Wang (2014), Hsu, Lin, and Vincent (2016).
2. Trading strategy: Neuhierl and Schlusche (2011), Metghalchi et al. (2012), Genre et al. (2013), Goyal and Wahal (2015).
3. Calendar effect: Sullivan, Timmermann, and White (2001), Hansen, Lunde, and Nason (2004), Coakley, Marzano, and Nankervis (2010).
4. Model performance: Hansen (2005), Hansen and Lunde (2005), Kao, Kuan, and Chen (2013).
5. Mutual/hedge fund: Romano and Wolf (2005), Chuang and Kuan (2010, 2017), Yen and Hsu (2010).

An Example: Hsu, Kuan, & Yen (2014)

- To identify outperforming Commodity Trading Advisor (CTA) funds, a subset of Macro hedge funds.
- Data: A total of 1050 funds during July 1994–June 2010
  - Following Kosowski, Naik, and Teo (2007), we exclude the first 12 months of data, so as to mitigate the incubation bias.
  - Small funds with assets less than $20 million are also excluded because they are not available to general investors.
- Fund performance measure: The t-ratio of the constant term (Alpha) of CAPM, the 4-factor model of Carhart (1997), and the 5-factor model of Fung and Hsieh (1997, 2001). An Alpha represents a portfolio's risk-adjusted excess return.

Sample: July 1998– June 2008

Nominal FWER(k)=5%										
		All funds			Discretionary			Systematic		
Model	Test	$k=1$	2	3	$k=1$	2	3	$k=1$	2	3
CAPM	Step-RC(k)	1	8	12	3	5	5	0	9	9
	Step-SPA(k)	1	8	12	3	5	5	0	9	9
4-factor	Step-RC(k)	0	0	0	0	5	7	0	0	3
	Step-SPA(k)	0	0	4	0	5	7	0	1	5
5-factor	Step-RC(k)	4	5	8	1	3	3	4	4	11
	Step-SPA(k)	4	5	10	1	3	3	4	9	16
All 3	Step-RC(k)	0	0	0	0	2	3	0	0	3
models	Step-SPA(k)	0	0	0	0	2	3	0	1	5

Note: There is a total of 77 funds, in which 14 are discretionary and 63 are systematic.

False Discovery Rate (FDR)

Controlling k (usually a small integer) false rejections may still be too stringent when m is very large. Instead, one may want to allow for a given proportion of false rejections among all rejections. Consider the false discovery proportion (FDP):

$$\text{FDP} = \frac{\text{Number of false rejections}}{\text{Number of total rejections}}.$$

It is common to control its expection, the false discovery rate (FDR):

$$\text{FDR} = \mathbb{E}(\text{FDP}).$$

Unlike FWER which is a type-I error probability, FDR is not. Nonetheless, FDR carries a probability interpretation (the Bayes posterior probability of nullness); see Efron and Hastie (2016).

The Benjamini and Hochberg (1995) Procedure

Let $p_{(1)} < p_{(2)} < \cdots < p_{(m)}$ denote the p-values of m tests arranged in an ascending order. The BH procedure finds

$$k^* = \max\left(k : p_{(k)} \leq \frac{k \times \theta}{m}\right),$$

where $0 < \theta < 1$ is a pre-specified number, and rejects the hypotheses corresponding to $p_{(1)}, \ldots, p_{(k^*)}$. This is analogous to the Holm procedure. In practice, it is common to set $\theta = 0.1$.

- Check the least significant $p_{(m)}$ and stop the procedure if $p_{(m)} \leq \theta$.
- Otherwise, proceed to check $p_{(m-1)}$ and stop the procedure if $p_{(m-1)} \leq (m-1)\theta/m$.
- This procedure continues till the first $p_{(k)}$ satisfies this inequality.

Benjamini-Hochberg FDR Control

When p-values (individual statistics) are independent of each other,

$$\text{FDR} = \pi_o \theta \leq \theta,$$

where $\pi_o = m_o/m$, with m_o the number of true nulls.

In a large-scale testing problem, most of the null hypotheses are true so that π_o is usually close to 1.

To improve the power of the BH procedure, Giglio, Liao, and Xiu (2021) suggest a screening scheme that drops obviously true nulls (i.e., very negative statistics) and consider only the remaining hypotheses. A smaller m leads to a larger p-value threshold and better test power. In particular, a hypotheses would be dropped if

$$t_i < -c \log(\log T) \sqrt{\log m}.$$

The Storey (2005) Procedure

Recall that BH's FDR control is FDR $= \pi_o \theta \leq \theta$, where $\pi_o = m_o/m$, yet the information about m_o is not utilized in the BH procedure. To this end, Storey (2005) estimates π_o using an estimator $\hat{\pi}(\lambda)$, where λ is the threshold defining the true nulls (details omitted). With the rejection threshold κ, the proposed estimator of FDR is:

$$\widehat{\text{FDR}}_\lambda(\kappa) = \frac{\hat{\pi}(\lambda) \cdot \kappa \cdot m}{\text{Number of } p_i \leq \kappa},$$

where the numerator is the estimated number of false rejections based on the assumption that true nulls follow a uniform distribution, and the denominator is the number of total rejections.

It can be seen that the Storey procedure in effect finds

$$h^* = \max\big(h : \widehat{\mathrm{FDR}}_\lambda(p_{(h)}) \le \theta\big),$$

where $\widehat{\mathrm{FDR}}_\lambda(p_{(h)}) = \hat{\pi}(\lambda) \cdot p_{(h)} \cdot m/h$. It follows that

$$h^* = \max\left(h : p_{(h)} \le \frac{h}{m}\theta \cdot \frac{1}{\hat{\pi}}\right),$$

where $h\theta/m$ is the probability threshold in the BH procedure. Storey's threshold is larger than BH's because $1/\hat{\pi} > 1$ and thus permits more rejections. This suggests that h^* must be greater than k^* in the BH procedure, rendering a more powerful test.

Note: The Storey's method is applicable when the nulls are equalities. If not, the p-values are not uniformly distributed and violate the underlying assumption. In this case, $\widehat{\mathrm{FDR}}$ becomes an invalid estimator.

Empirical Applications

1. Mutual fund and manager performance: Fama & French (2010); Barras, Scaillet, & Wermers (2010, 2019); Cuthberton, Nitzsche, & O'Sullivan (2010, 2012); Ferson & Chen (2021); Wang, Han, & Tong (2023); Huang, Lee, & Rennie (2026)
2. Trading rules: Bajgrowicz & Scaillet (2012); Harvey & Liu (2015); Chen & Han (2019); Sermpinis et al. (2021)
3. Investment Strategies: Harvey & Liu (2020)
4. Ecology: Betts et al. (2017)

Concluding Remarks

1. In large-scale, multiple testing problems, it is crucial to control a proper measure of familywise error, e.g. FWER, FWER(k), FDR, etc. The choice of such error measure ought to be context dependent.
2. It is also important to take into account the dependence among individual statistics in constructing a multiple testing procedure. For example, the BH procedure is applicable when the statistics are independent or have some form of positive correlations (e.g., positive regression dependency on a subset). See Benjamini & Yekutieli (2001) for more results.
3. Many empirical applications may be re-studied using these multiple testing methods; Giglio, Liao, & Xiu (2021) provide a good example.